Big Data as a Governance Mechanism

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This study empirically investigates two effects of alternative data availability: stock price informativeness and its disciplining effect on managers' actions. Recent computing advancements have enabled technology companies to collect real-time, granular indicators of fundamentals to sell to investment professionals. These data include consumer transactions and satellite images. The introduction of these data increases price informativeness through decreased information acquisition costs, particularly in firms in which sophisticated investors have higher incentives to uncover information. I document two effects on managers. First, managers reduce their opportunistic trading. Second, investment efficiency increases, consistent with price informativeness improving managers' incentives to invest and divest efficiently. (*JEL* G14, G12, G23, G34, O16, M12)

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In this study, I examine whether the availability of alternative data improves price informativeness and helps discipline corporate managers. Price informativeness and the allocation of information in an economy are important because they have the potential to affect managers' actions (e.g., Fishman and Hagerty 1989; Holmstrom and Tirole 1993; Brandenburger and Polak 1996). It can be empirically challenging to assess whether managers take different

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actions when prices become more informative because of the endogenous nature of price informativeness and corporate disclosure. For example, a manager might choose to make less informative disclosures to benefit from personal trades in the firm's stock. Similarly, he might disclose less when he chooses less efficient investments. To address the challenge of empirically studying this relation, I first test for an increase in price informativeness that results from technological innovations exogenous to the firm's managers. I then evaluate the disciplining effects of this improved informativeness on managers' opportunistic trading and real investment decisions.

I test for a change in price informativeness using the growth in alternative data sets, some of which are referred to as "big data." Alternative data are defined as data sets that are "not from a financial statement or report" (Quinlan and Associates 2017). In recent years, the proliferation of mobile devices, low-cost sensors, and other technologies has reduced data-gathering costs, leading to the birth of multiple start-ups that collect these alternative data. These data include point-of-sale transactions, satellite images, and clickstream data, and they are different from traditional sources of information (e.g., financial information from company filings, investor presentations, and analyst reports) in that they are granular, real-time data that are not derived from firm disclosures. The availability of these third-party data sets has reduced investors' costs of acquiring information, such that investment professionals have begun to use these data in investment strategies (Bank of America Merrill Lynch 2016). Despite the increased use of these alternative data sets, there is limited or no research on their consequences for capital markets and managers' actions.

First, I examine the link between alternative data and improved price informativeness. Noisy rational expectations models predict that, when information acquisition costs decrease, the informational efficiency of stock price increases (Grossman and Stiglitz 1980; Diamond and Verrecchia 1981; Verrecchia 1982). In these models, prices do not perfectly convey the private signals of informed investors; consequently, a decrease in the cost of information acquisition improves price informativeness, defined as the correlation between price and fundamental information. However, it is less clear whether traders' acquisition of the data can lead to an improvement in long-run price informativeness (McNichols and Trueman 1994). The data sets presumably contain short-horizon indicators of fundamentals (e.g., consumer transactions that have occurred but have not yet been announced by the firm). Therefore, it is an open question whether the availability of alternative data can improve the incorporation of long-horizon (i.e., 1-year-ahead) earnings into prices.

To assess whether alternative data are indeed informative, I obtain access to two alternative data sources. The first data source contains consumer transactions from a marketing analytics platform built on a large panel of consumer-browsing data, which are passively collected from users that have installed antivirus software and sold to active portfolio managers. For example, the data include unique checkout transactions completed at consumer-facing firms with an online presence (e.g., macys.com). The second data source is a satellite image data partner that provides normalized car counts in parking lots of retailers. These car counts map to consumer transactions in stores and are relevant for firms with a retail store presence. These data cover 266 firms from 2014 to 2016. I show that aggregated signals from these data sets have predictive power for revenue and earnings that are not yet announced, and the data can predict announcement returns.[1] After verifying that these data sources contain incremental information content, I validate that investors use the data by showing that price reactions to earnings announcements are muted after alternative data from these data sources are available in June of 2014. Despite the prohibitive costs of these data sets (i.e., hundreds of thousands of dollars), the availability of alternative data results in a measurable increase in short-run price informativeness. Inferences are based on a difference-in-differences research design comparing the 266 firms covered by these alternative data sets to a group of matched firms that are economically similar but do not have much data coverage.

Following these validation tests, I test for an increase in long-run price informativeness. The richness and granularity of the alternative data contain information that is typically not publicly disclosed by the manager, and this superior information can help investors incorporate fundamental information related to longer-term performance into prices.[2] I find evidence that, for firms affected by alternative data, current returns contain more information about future earnings. Cross-sectional tests find that the short-run and long-run effects are stronger in firms for which sophisticated investors have the highest incentives to uncover information (i.e., firms that sell discretionary consumer products and services, firms with higher market-to-book ratios, and more liquid firms). The results seem to be driven by sophisticated investors who presumably acquire these alternative data sets.

After finding the improvement in long-run price informativeness after alternative data are available, I then focus on two potential effects on the manager. For the first effect, I investigate whether the investor's use of alternative data reduces the manager's opportunity to trade on his private information about future earnings. The literature on insider trading has found that managers exploit their superior information for personal trading gains (Piotroski and Roulstone 2005; Rogers 2008). When prices reflect future

[1] Using both data sources, I find that these alternative data predict revenue and earnings that will be announced after the end of the quarter. They also predict announcement period returns when those revenue and earnings numbers are released. A long-short trading strategy earns 1.4% to 2.0% in abnormal returns in the 11-day [-5, +5] window around the earnings announcement (see the Internet Appendix).

[2] This assumption is consistent with Froot et al.'s (2017) finding that managers do not disclose all of their private information. Similar to that used by Froot et al. (2017), the private information proxy in this paper is based on big data. However, Froot et al. (2017) do not test the impacts of these data on price informativeness or on managers' actions, which are the focus of my paper.

earnings more quickly and completely, the manager has less of an opportunity to extract rents by trading on his private information. Consistent with this hypothesis, I find that insiders of firms affected by alternative data are less likely to purchase shares ahead of positive future earnings innovations. Furthermore, when insiders do trade, I find that the positive relation between insider trades and future earnings innovations is attenuated after alternative data are available. This evidence suggests that managers reduce the exploitation of their private information about future earnings through personal trades when prices reflect information from alternative data.

The second managerial action effect I investigate is whether alternative data availability disciplines the manager to make better real investment decisions. Agency problems, which result from the separation of ownership and control, have been shown in several papers to induce empire building or overinvestment of free cash flow (Jensen 1986; Harford 1999; Bates 2005; Richardson 2006). Furthermore, concerns about reputation and reluctance to take action (i.e., the quiet life hypothesis) hinder the manager's discontinuation of underperforming businesses (Kanodia et al. 1989; Boot 1992; Bertrand and Mullainathan 2003). When investment opportunities are declining, the optimal firm response is to curtail investment Wurgler (2000). Therefore, managers' incentives to expand the size of the firm (i.e., empire building) instead of closing businesses (i.e., reputation and the quiet life) are misaligned with shareholders' when investment opportunities are declining.

In my setting, the documented increase in long-run price informativeness is consistent with alternative data providing information about future profitability, whether that profitability is related to assets in place or expected future investment opportunities. With respect to assets in place, alternative data might reveal granular information about which businesses should be closed. With respect to investment opportunities, alternative data might reveal superior information about which businesses to expand. I acknowledge that I cannot directly observe whether corporate managers are aware of alternative data's effect on sophisticated investor behavior and prices. Therefore, my tests are joint tests of this awareness and the effect on firm choices. Following Wurgler (2000), I define the level of investment efficiency as the responsiveness of the firm to investment opportunities (i.e., increasing investment when opportunities are expanding and, conversely, decreasing investment when opportunities are deteriorating). I test for changes in this responsiveness and find that the introduction of alternative data to the market is associated with a greater sensitivity of investment to deteriorating opportunities. Consistent with prior research, I do not find the symmetric effect when investment opportunities are expanding (Wurgler 2000; Bushman et al. 2011). In additional tests of investment efficiency, I find that the excess returns to announcements of discontinued operations are higher after alternative data availability.

My study makes two main contributions. First, I contribute to the growing literature on the impact of technology on capital markets. Recent papers have documented the capital market effects of multiple technological innovations, including algorithmic trading (Hendershott et al. 2011), high-frequency trading (a form of algorithmic trading studied in Brogaard et al. 2014), and robo-journalism (Blankespoor et al. 2018). My paper examines another technology-related impact on the capital markets: the impact of the use of alternative data in asset management. Asset managers invested an estimated $4 billion into alternative data in 2017 (Marenzi 2017), but to date there has been little research in this area. My study builds on the finding in prior research that alternative data sources can predict earnings and revenue (e.g., Froot et al. 2017) and investigates the fundamental question of how information acquisition costs affect price informativeness in a current setting, where the cost reduction occurs for a set of sophisticated investors. Prior studies largely focus on cost reductions for a broad set of investors, although these reductions may benefit sophisticated investors more so than others (Blankespoor et al. 2014). I provide new insights into the effect of concentrated information, as the cost reduction I study occurs only for this subset of investors, given that costs of these alternative data sets are prohibitive for other investors.

The second contribution of my study is to investigate how capital market forces can reduce agency costs. Agency problems motivate managers to exploit their private information through personal trades and to make inefficient investment decisions. I document that increased price informativeness can have a disciplining effect on corporate managers, consistent with theories of the effect of price on managerial actions (e.g., Holmstrom and Tirole 1993; Baiman and Verrecchia 1996; Baker et al. 2003; Jensen 2005; Polk and Sapienza 2009). Empirically testing these theories is challenging: price informativeness and corporate governance are endogenously determined because of potential reverse causality or omitted variables. Reverse causality can be especially difficult to tackle, as governance issues are typically persistent in the time series within a firm. With my setting, I study an increase in price informativeness that is exogenous to the manager's choices. Prior literature has studied the effect of disclosure choices and regulation on investment efficiency (e.g., Biddle and Hilary 2006; Hope and Thomas 2008; Biddle et al. 2009; Bushman et al. 2011; Shroff et al. 2014). I contribute to this literature by studying the effect of alternative data, which are distinct from the firm's disclosure choices and regulatory changes, on the manager's actions.

1. Background and Development of Hypotheses

1.1 Institutional background

A consensual definition of big data is "the information asset characterized by such a high volume, velocity and variety to require specific technology and analytical methods for its transformation into value" (De Mauro et al. 2016, p. 1). The recent improvements in data storage, cloud computing, and machine learning have gradually reduced costs of gathering data, leading to the birth of multiple startups that collect data.[3] Some third-party vendors use high-resolution satellite images to count the number of cars in parking lots, while others extract information from consumers' online activity or measure foot traffic in stores. Another set of companies uses credit card transactions to understand where consumers are spending their money. Data are generated from individuals, business transactions, and sensors, all of which have a heavy emphasis on consumer "footprints." In contrast, fewer data are available on firms that do not directly sell their products to consumers.

The introduction of these third-party vendors represents a sharp, observable reduction in the cost of information acquisition for a subset of investors. Fundamental, quantitative, and other active portfolio managers are equipped to find new sources of data, develop and test hypotheses based on these data, and trade according to insights from these hypotheses.[4] An estimated \$4 billion was spent in 2017 on the use of alternative data in investment strategies, and this number will grow to \$7 billion in 2020 (Marenzi 2017). In equilibrium, the money the industry invests in these data sources is a "shadow price" for the value of the data. Prices are high because of high demand, low price elasticity, and the desire to keep data sets relatively exclusive (Bank of America Merrill Lynch 2016).

1.2 Development of hypotheses

1.2.1 Price informativeness. My first hypothesis, stated below in alternative form, is based on the conjecture that the availability of alternative data reduces the cost of information acquisition.

H1. The availability of alternative data increases price informativeness.

The efficient market hypothesis states that stock prices reflect all available information (Fama 1970). However, this hypothesis relies on information being costless. In a market with costly information acquisition, in order to compensate informed investors for incurring the costs to acquire information, their trades must occur at prices that do not fully reflect the information signal; otherwise, there would be no incentive for them to acquire the information in the first place (Grossman and Stiglitz 1980). If I do not detect a measurable increase in price

[3] "Historically big data was out of reach for investment managers, given its complexity and unstructured nature. But recently [there has been] a significant increase in entrepreneurial based technology startups […] The advancement in computational power and cloud computing environment is also reducing the entry barriers in this space" (Deutsche Bank 2016).

[4] A Barclays survey found that 24% of discretionary hedge funds use alternative data (Eagle Alpha 2017). Eagle Alpha, a provider of research and aggregator of alternative data sets, also estimates that 150 firms have at least one person dedicated full-time to alternative data. Two investment companies at the forefront of the push toward alternative data are Point72 and BlueMountain (Foxman 2017). Quantitative funds, such as Two Sigma, WorldQuant, Citadel, and Blackrock, analyze massive amounts of often unstructured data to make investment decisions (Kishan 2017;Thomas 2017b). In 2017, quantitative-focused hedge funds held more than 30% of all hedge fund assets and were responsible for 27% of U.S. stock trades, compared to 14% in 2013 (The Tabb Group 2017; Zuckerman and Hope 2017).

informativeness, (1) markets are strong-form efficient, (2) the reduction to the cost of acquiring information was not great enough for me to detect an effect, or (3) the data are uninformative.[5]

First, I discuss (1). Before the availability of alternative data, this information was known by some market participants (e.g., privately informed corporate managers) and the information could be obtained by sophisticated investors willing to incur high costs (e.g., hiring a person to physically count foot traffic with clickers).[6] I assume that the availability of alternative data, under semi-strong form market efficiency, allows market participants to acquire this information at a lower cost and with greater precision.

I rely on models in which prices imperfectly convey information in the private signal (e.g., Grossman and Stiglitz 1980; Verrecchia 1982). A decrease in the cost of information acquisition leads more investors to choose to become informed, which increases the price system's informativeness (Grossman and Stiglitz 1980). In Verrecchia (1982), a model in which investors endogenously choose the level of precision of information to acquire, price informativeness increases with a reduction in the cost of information acquisition, because investors acquire more precise information.

Regarding (2), the effect may be empirically undetectable, despite the predictions of this class of models. It is difficult to glean insights from the data (e.g., analysis requires a data science team) and data sets are expensive (e.g., hundreds of thousands of dollars). Therefore, the cost reduction affects only a subset of investors. Furthermore, the reduction in the cost of acquiring and implementing a particular signal could affect traders' acquisition of other signals, depending on the joint distribution of signals, payoffs, and equilibrium prices (Admati and Pfleiderer 1987). Strategic complementarities in trading decisions can lead to excess volatility in trading and prices if traders herd on information (Froot et al. 1992; Veldkamp 2006a, 2006b; Amador and Weill 2010; Garciá and Strobl 2011).

My empirical validation tests address (3) that the data might be uninformative. Imprecision in the data could have an ambiguous effect on price informativeness (Brunnermeier 2005). Traders with alternative data could distort prices and increase volatility.[7] Using methods similar to those that

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5 I do not address the effects on liquidity in this paper. It is unclear ex ante whether liquidity would increase or decrease in the post-data availability period. Kyle (1985) predicts an increased price impact when the informed trader has a more precise information signal, but models, such as Holden and Subrahmanyam (1992), extend the Kyle (1985) model to multiple agents and show that with multiple informed traders and Cournot-like competition between them, the aggressive trading increases market depth. Liquidity could improve when there is less information asymmetry between the market maker and the manager, because the market maker extracts less information from net demand, which makes price less sensitive to net demand (Baiman and Verrecchia 1996).

6 With respect to satellite imagery, "alternative data approaches are faster and more comprehensive than physical [clicker] counts, leading to an information advantage over the old-school approach—even though the data sets were measuring similar consumer activities" (Deloitte 2017).

7 Although the extensive resources institutional investors devote to securing and analyzing these data suggest that they are likely useful to trading decisions, critics argue that the data are often inaccurate and are misleading

portfolio managers might use, I find that the data predict revenue and earnings, and a long-short strategy earns 2% in abnormal returns in the 11 days around the earnings announcement (see the Internet Appendix).

While the data provide information about current quarter earnings, it is unclear whether they can also provide information about longer-term fundamentals. For example, online consumer transaction data can be matched to individual products, and investors can understand the growth prospects and competitive positioning of firms at a granular level.[8] However, it might be challenging to use the data in this way, and it is unclear whether the data have predictive value for longer-term earnings. Dugast and Foucault (2018) theoretically show that the effect of reduced costs of "raw" (i.e., alternative data) signals on long-run price informativeness is ambiguous, because a decrease in the cost of producing this "raw" signal could increase or reduce demand for the processed (i.e., more informative and less noisy) signal. Furthermore, prices might not incorporate information about future earnings, because a trader that receives a signal about long-run value may choose not to trade until a later date, depending on the relative profitability of trading earlier versus later (McNichols and Trueman 1994). Therefore, it is an empirical question whether prices will contain more information about future (1-year-ahead) earnings after alternative data are available. Next, I discuss two managerial actions that can be disciplined by this increase in long-run price informativeness.

1.2.2 Effects of price informativeness on corporate managers' personal trades. The first disciplining effect that I investigate is whether increased price informativeness constrains managers to extract fewer information rents from shareholders.

H2a. The availability of alternative data reduces the magnitude of the relation between insiders' trades and future unexpected earnings, conditional on the decision to trade.

H2b. The availability of alternative data reduces the propensity of insiders to purchase (sell) shares ahead of positive (negative) future unexpected earnings.

Prior literature has shown that managers' stock purchases and sales are related to their private information about the firm's future earnings. When prices

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traders and increasing market volatility. For example, the data provided misleading predictions about Netflix's earnings in October 2016: "On Oct. 5, Earnest [a card spending data company] sent a note to clients saying Netflix's paid domestic streaming subscriber numbers were tracking below consensus at the end of the third quarter. Short interest climbed leading into the earnings report, and the shares shot up 19% on Oct. 18 after Netflix beat consensus estimates on that figure" (Gottfried 2017).

8 "Consumer transaction data can also be used by long-term investors e.g. to evaluate online/offline shopping habits, product success, brand stability, stage of product adoption, demographics of customer base, and the temporal impacts of promotional campaigns" (Eagle Alpha 2017). "For example, if you look at customer profiles for two competing products and find that one skews younger, richer, and more urban? That product would probably be better positioned for the future than another one whose audience is mainly retirees and lower income. A retailer may say they're going to target a particular audience and you can overlay transaction and demographic data to determine whether or not they are successful in doing that" (Thomas 2017a).

reflect information about future earnings to a greater extent, there is less of an opportunity for the manager to trade on his private information about future earnings. Piotroski and Roulstone (2005) find that firms with better information environments, as proxied by firm size and analyst coverage, have insider trades that are less related to future earnings innovations. An improvement in the information environments of firms covered by alternative data could constrain the ability of insiders to trade on their private information. Managers can extract rents from shareholders by trading in anticipation of future earnings, and this ability to extract rents is inversely related to price informativeness. I test whether increased price informativeness disciplines the manager's decision to trade and the directional magnitude of trades. If the more informative trades are disciplined, then conditional on the decision to trade, the informativeness of the remaining trades might be lower.

1.2.3 Real effects of price informativeness: Investment efficiency. My next hypothesis is related to the misalignment of incentives with respect to the manager's real investment decisions. I test whether alternative data availability affects the efficiency of real decisions, but I caution that this real effect may take longer to appear. I can only document whether investment efficiency has increased in the few years since alternative data have been available, and if I do not detect an effect, (1) alternative data use by investors has no impact on real efficiency or (2) the effect has yet to occur in my sample.

H3. The availability of alternative data improves investment efficiency.

Prior literature has found weak evidence that traditionally defined governance structures explain the mitigation of overinvestment (Richardson 2006). Among other considerations (e.g., reputational concerns, expectations of selling his shares in the future), the manager cares about potential price changes because price is weighted in his compensation contract. The concern about stock price can induce inefficient investment decisions (Stein 1989). I focus on informationally efficient prices as a governance mechanism that can discipline managers to invest efficiently. An increase in the amount of information in stock price is hypothesized to improve these incentives (Holmstrom and Tirole 1993; Baiman and Verrecchia 1996; Brandenburger and Polak 1996; Edmans 2009). Alternative data, if they reveal information about future earnings (H1), may reveal the specific information about current businesses and future investment opportunities necessary for the market to assess investment and divestment decisions. For example, the granular transaction-level data can reveal which businesses are performing well and which are underperforming. I assume this information is the manager's private information, which previously could not be credibly communicated (e.g., Kanodia and Lee 1998) or was costly to disclose (e.g., Verrecchia 1983). This information reveals into which businesses the firm should expand investment and into which it should decrease investment. Decreasing investment can be as extensive as reducing investment beyond the level of investment required to maintain assets in place (e.g., at an extreme, closing businesses).

Detecting increased investment efficiency after alternative data are available is consistent with managers being disciplined by the threat of incorporation of this information into prices.[9] Managers' actions are revealed in the data in a timely and granular manner. For example, the value implications of a manager's decision to open a new store or invest in research and development (R&D) to develop new products are apparent in the satellite image or consumer transactions data sets in real time and at the store level and at the product level. With fewer confounding events when the action and the data outcome are close in time, a sophisticated investor with real-time data should be better able to monitor the manager's actions. While alternative data can reveal information about both positive and negative performance, there is an asymmetric misalignment of incentives when businesses are underperforming compared to when they are performing well. Empire building tendencies Jensen (1986) and the reluctance to divest underperforming businesses (Kanodia et al. 1989; Boot 1992; Weisbach 1995; Bertrand and Mullainathan 2003) are more problematic when those businesses are underperforming.

My tests follow the definition of investment efficiency in Wurgler (2000), which is the responsiveness of investment to improving or deteriorating investment opportunities. Given that the misalignment of managerial incentives with those of shareholders is more severe when investment opportunities are deteriorating, I focus on whether the availability of alternative data to the market curbs investment in declining industries. In a cross-country study, Wurgler (2000) finds that investor rights, which provide managers with strong incentives to maximize firm value, are associated with keeping investment out of declining industries. This finding supports the ability of minority investors to exert pressure on managers to invest free cash flow efficiently, consistent with Jensen's (1986) free cash flow theory. Similarly, Bushman and Smith (2001) find that timely loss recognition has the same disciplining effect on investment in declining industries, consistent with shareholders and lenders being able to respond quickly to a deterioration in the firm's profitability or financial condition.

These hypothesized effects on managers are the result of the presumed users of alternative data, sophisticated investors, including hedge funds that hedge a long position by taking short positions (see Section IA.2 of the Internet Appendix). Short sellers been shown in prior literature to exert pressure on

[9] This channel is similar to the disciplining channel of a large shareholder with a credible threat of exit (Admati and Pfleiderer 2009). An alternative disciplining channel is behind-the-scenes intervention ("voice") (e.g., McCahery et al. 2016) or hedge fund activism (e.g., Bebchuk et al. 2015). Although I do not attempt to rule out the potential acquisition of alternative data by activist investors or investors who use "voice" to intervene, I note that the disciplining channel of the threat of incorporation into price is plausible, given that alternative data are marketed to quantitative-focused hedge funds. Investors that are more concerned about liquidity, such as quantitative-focused hedge funds, use "voice" less intensively (McCahery et al. 2016).

managers by impounding negative information into stock prices quickly. The threat of sophisticated investors responding quickly to managers' investment decisions or deterioration in firm profitability disciplines managers' actions, including inefficient investment and divestment decisions. Multiple studies find that short sellers increase the informational efficiency of prices (e.g., Diamond and Verrecchia 1987; Dechow et al. 2001; Karpoff and Lou 2010; Boehmer and Wu 2012) and discipline managers' behavior (e.g., Goldstein and Guembel 2008; Massa et al. 2015; Grullon et al. 2015; Fang et al. 2016).

2. Data

I use two alternative data sources in my analyses. While I clearly cannot capture the entire corpus of data utilized by active portfolio managers, I am able to document that certain important data sets are relevant and have implications for price informativeness. In the Internet Appendix, I validate that these data sets have potential predictive ability for the future earnings of consumer-focused companies. Because of the nature of the data collection process, most of the data sets available study consumer behavior.[10] I describe in more detail the data sets I use next.

2.1 Source 1: Online consumer transactions

The first data source I use contains browsing data for a panel of consumers, and its panel is orders of magnitude larger than those of web traffic data previously available through PC Data, Nielsen/Netratings, and comScore Media Metrix.[11] The data set classifies clicks, browsing sessions, and unique devices into event categories, including "conversion" (purchase) and "startcheckout" (user began the checkout process). The panel begins in early 2014, so the first fiscal quarter end with full data is either March 31, 2014 or June 30, 2014, for almost all Web sites.

2.2 Source 2: Satellite image data for car counts

The second data source is a geoanalytics platform that provides access to and analysis on geospatial data. Using image processing, machine learning, and cloud computing, this company partners with satellite imagery providers to

[10] A Q4 2016 Data Sets Market Survey confirms this assumption, as almost all of their data sets are marked as useful for the consumer sector, and data sets useful for firms in other sectors are sparse (Bank of America Merrill Lynch 2016). A May 11, 2017 email update from Eagle Alpha confirms that the three most popular alternative data sets (out of the 486 they offer) are all related to consumer transactions (e.g., credit card transactions and email receipts). The Consumer Discretionary sector has 3.4x the number of data sets compared with the other 10 sectors studied by Eagle Alpha (Eagle Alpha 2017). I also conducted interviews with multiple hedge fund analysts and other industry experts, who confirmed this assumption.

[11] Other data sources with consumer transactions include credit card transaction data providers, such as Yodlee and Cardlytics. These companies sell credit card transaction data from consumers who have opted in to provide their anonymized transaction data to these companies in exchange for services from financial institutions, such as account aggregation and analytics. Some small businesses are also included in these credit card transactions, and I discuss these implications in Section 3.1.

understand where consumers are going. For example, its car park observations are normalized to discard employee cars and adjust for seasonality, and they are useful for understanding consumer shopping behavior on a daily basis. The data are aggregated at the ticker level, and they were sold to clients beginning in 2014.

2.3 Limitations

The inferences of my study are limited because I cannot directly observe the clients of these data providers, when they purchased data, and whether or not they use these specific data sets in their investment decisions. However, I confirm the assumption that the purchasers of alternative data are sophisticated investors, many of which manage short positions by (1) conducting interviews with data providers and industry professionals, (2) validating the increase in short seller activity for my covered firms in Section IA.2 of the Internet Appendix, and (3) conducting cross-sectional analyses discussed in Section 3.1.1. Furthermore, I do not have direct evidence of whether corporate managers are aware of investors' use of alternative data. Consequently, my research design focuses on testing for outcomes of these connections; I examine firms covered by the data providers in my sample, and I carefully form a control group of firms that does not have much data coverage but is economically linked to the covered sample.

Another limitation of my setting is that I cannot observe the private information of the manager. I can test whether the data provide foreknowledge of publicly announced revenue and earnings, but I can only provide indirect evidence consistent with the conjecture that the data contain information that was formerly the manager's private information. I provide such evidence in Section 3.4.

3. Methodological Approach and Empirical Results

3.1 Sample construction

My sample consists of firms whose data are gathered and released by the data providers and a control group of firms whose data are not in the data providers' data sets. The data providers have given me access to a subset of their data, but their raw data sold to sophisticated investors cover a larger set of firms. To identify which firms are truly "covered" by alternative data, I assume that the raw data cover all firms that sell products (i.e., sell products to the same types of customers) similar to those of the firms in the subset provided to me. Similar consumers purchase products from firms that sell similar products. I use the Hoberg-Phillips text-based industry classification (TNIC) system, which allows me to identify the 10 closest peers of each firm in the data sets from data sources 1 and 2, based on the textual similarity of their 10-K product descriptions (Hoberg and Phillips 2010).[12] I use each firm's TNIC peers rather than the

[12] See Section IA.1.3 of the Internet Appendix for a depiction of this procedure. In Section IA.1.3, I also provide results of cross-validation procedures to assess the false-negative rate of this procedure to identify truly covered

NAICS or SIC industry peers, because the NAICS and SIC classifications are based on production processes, whereas the TNIC classifications are based on product market similarities, which map more closely to the similarities in firms I aim to capture. Covered firms are the union of the set of firms inferred to be covered and the set of firms in the subset of data provided to me.

Next, I identify a set of potential control firms. The difference-in-differences empirical design attempts to establish the effect of alternative data coverage, relative to the counterfactual outcome of no coverage by alternative data. The true counterfactual outcome is unobservable, so I assess the change in outcome variables for the covered firms relative to that of a set of control firms to represent the inferred counterfactual outcome. The goal is to ensure that these control firms are relatively less affected by alternative data coverage but are otherwise similar to the covered firms. To identify firms that are affected by the same economic factors as my covered firms, I rely on prior literature showing supplier-customer industries' correlated economic fundamentals and investors' limited incorporation of customer industries' fundamental information into the returns of firms in supplier industries (Menzly and Ozbas 2010). I also assume that other information environment variables unrelated to alternative data coverage remain constant between the two groups of firms. I use the 2014 input-output flow tables from the Bureau of Economic Analysis (BEA) Web site to identify firms in BEA industries that supply at least 5% of their output or whose supply comprises at least 5% of the input to the BEA industries of firms in my aggregate data (see the Internet Appendix Figure IA-3). I exclude financials and utilities in this process. While the potential control firms are economically similar to my Covered firms, they are not as affected by alternative data coverage, due to the difficulty of incorporating information about customer firms into supplier firms' prices. I note that my selection of these economically related firms results in understating the effect of alternative data, because alternative data availability likely affects these potential control firms as well.[13]

This procedure results in 1,932 unique potential control firms for the 266 covered firms. To ensure that the two sets of firms are affected by similar information environments, I mitigate differences in firm size across the two groups. Specifically, I match each covered firm to a control firm that is closest in size using an optimal matching algorithm that minimizes the absolute distance across all matched pairs (Ho et al. 2011). This procedure results in 266 matched pairs of firms. Table 1 reports the descriptive statistics.

firms. The size of the firms is the main difference between the 120 firms in the subset provided to me and the 146 firms inferred to be covered. The inferred covered firms are smaller (median \$1.5 billion market cap) relative to the firms in the subset provided to me (median \$3.3 billion market cap).

13 In addition, credit card transactions, which include some small businesses, can be used to understand the economic fundamentals of some potential control firms, for control firms that sell products to small businesses.

Table 1
Descriptive statistics of matched pairs

	Covered firms		Control firms	
	Mean	Median	Mean	Median
Market cap	11,078.52	2,192.36	13,574.97	2,047.00
Size	20,001.30	1,646.43	12,619.10	1,811.23
Analysts	6.10	5.00	4.98	3.00
BTM	0.36	0.30	0.36	0.29
Leverage	2.35	1.22	1.93	1.05
Beta	1.19	1.14	1.24	1.14

This table presents descriptive statistics, measured at the last fiscal quarter end on or before June 30, 2014, for the 266 matched pairs of firms (532 firms total) in the covered and control groups. Table A1 defines all variables. Descriptive statistics are presented for unlogged *Size* (total assets) and unlogged *Analysts*.

For unbiased estimation, the identifying assumption of the difference-in-differences design is that the covered and control firms would have followed parallel trends under the counterfactual condition that the covered firms were not covered by alternative data. This assumption is impossible to test directly, which is a limitation of all studies with this type of design (Imbens and Wooldridge 2009). However, I can test whether the pre-period trends in the outcome variables are similar between covered and control firms. I test whether pre-period trends in the outcome variables are similar and exclude from the analyses matched pairs whose pre-period trends are dissimilar.[14] Analyses of pre-period trends find no significant differences in trends in variables of interest between covered and control firms in the pre-period (see the Internet Appendix). I select 2009 as the pre-period start year, to avoid including the financial crisis of 2007–2008.

These assignments of firms to the covered and control samples attempt to mitigate between-group differential changes in underlying economics and the information environment that are *unrelated* to the availability of alternative data. The control firms sell merchandise to other businesses (e.g., wholesale trade firms comprise a large portion of the control sample), and they are often upstream the supply chain relative to the covered firms that sell products directly to consumers. While aggregate consumer demand affects both groups of firms, the consumer transactions at a granular level are more useful in understanding firm value for the covered sample than for the control sample.

3.1.1 Cross-sectional variables. I supplement my main analyses by exploiting cross-sectional differences within the Covered sample. I use four cross-sectional variables related to investors' ex ante incentives to uncover information to profit from their information acquisition activities. The first two

14 Specifically, using only the pre-period observations, I calculate dfbeta for each matched pair. Doing so reflects how influential each matched pair is for the dissimilarity in pre-period trends. Before conducting each analysis, I iteratively delete matched pairs (usually at most 1 or 2 pairs are deleted), which contribute most to the deviation in parallel trends in the pre-period. I exclude the observations associated with these firms to ensure that there are no economically significant differences in pre-period trends across the two groups. This procedure occurs before I execute each analysis.

variables are related to the size of the profit opportunity, and the latter two variables are related to the constraints on investors' extraction of this opportunity.

First, as a proxy for a firm's exposure to information shocks, I use membership in industries with a high future total addressable market (*High_TAM*). These industries sell discretionary consumer products and services (i.e., apparel, restaurants, travel, and auto sales). For *High_TAM* firms, each data point (e.g., each consumer transaction) represents a large potential growth opportunity resulting in a large profit for a speculative trader. This variable is based on conversations with industry professionals. Second, as a proxy for a firm's sensitivity of future earnings to current earnings, I use the market-to-book ratio (*High_MtoB*). Prior literature has shown that earnings of firms with higher expected future earnings have a higher valuation impact (Kormendi and Lipe 1987). Third, as a proxy for low constraints to traders' extraction of profits, I use the Amihud (2002) illiquidity measure (*High_liq1*). Fourth, I use a related liquidity proxy, dollar trading volume (*High_liq2*).[15] I use these liquidity measures as proxies for the degree to which sophisticated traders' insights are actionable; a separate literature parses liquidity into adverse selection and noninformational components (Bharath et al. 2009).

3.2 The effect on price informativeness

3.2.1 Short-run price informativeness. I define short-run price informativeness as the degree to which prices reflect contemporaneous cash flows. Specifically, I validate that pre-earnings announcement prices reflect more current-period fundamental information that is released at the earnings announcement. Several models predict lower announcement price reactions when private information is gathered in anticipation of a public announcement (e.g., Kim and Verrecchia 1991; Demski and Feltham 1994; McNichols and Trueman 1994).

I study the information content of the news disclosed at the earnings announcement, operationalized as the market's assessment of the future cash flow implications of each unit of unexpected earnings (Collins and Kothari 1989). To control for all public information, it is important to test for changes in the slope coefficient of returns and unexpected earnings, rather than changes in the level of returns measured at the earnings announcement.[16] The advantage

15 Informed traders optimally withhold some information from price if the price impact of trading is high (Kyle 1985), so these final two cross-sectional variables are related. In a related study, Farboodi et al. (2017) find that big data growth generates improved efficiency for large firms vis-à-vis small firms. My cross-sectional results are complementary to theirs.

16 The announcement period return is a proxy for the amount of incremental value-relevant information about contemporaneous cash flows that is revealed at the earnings announcement. However, higher absolute announcement period return levels do not indicate that short-term prices are necessarily less informative. Industry reports have found that retailers' announcement returns are slightly higher in absolute terms, in recent years, and they claim that there is still alpha to be captured trading ahead of earnings announcements (Eagle Alpha 2017). My results are consistent with their findings, as I find slightly higher absolute earnings announcement returns in untabulated analyses. These results are concentrated in small absolute earnings surprise announcements (β_3),

of calculating unexpected earnings relative to the analyst consensus is that the analyst information set presumably includes all public information. Given that sell-side analysts likely do not purchase alternative data, due to the high costs of these data sets, the relation between the absolute value of unexpected earnings and announcement period absolute returns denotes the relative pre-earnings announcement information sets of investors who purchase the data and those that do not. The incremental information content of each unit of unexpected earnings will necessarily be smaller when prices impound more private information about not-yet-announced earnings (Skinner 1990).

Model 1 assesses whether the absolute announcement return per unit of unexpected earnings has decreased. The *Post* indicator variable equals 1 after the data providers begin to sell the data to portfolio managers, and *Covered* is an indicator variable equal to 1 for firms covered by data providers. Specifically, *Post* equals 1 when the fiscal quarter end is on or after June 30, 2014.[17] The sample consists of all firms covered by the data ($Covered=1$) and all matched control firms ($Covered=0$):

$$\begin{aligned} Abs\_AR[0,+2] = {} & \beta_0 + \beta_1 Post + \beta_2 Covered + \beta_3 (Post \times Covered) \qquad (1) \\ & + \beta_4 Abs\_UE + \beta_5 (Abs\_UE \times Post) + \beta_6 (Abs\_UE \times Covered) \\ & + \boldsymbol{\beta_7} (Post \times Covered \times Abs\_UE) + \Sigma \beta_k Controls_k \\ & + \Sigma \beta_k (Abs\_UE \times Controls_k) + \Sigma \beta_j Industry_j \\ & + \Sigma \beta_j Industry_j \times Abs\_UE + \varepsilon, \end{aligned}$$

where the dependent variable is the decile ranking of the absolute value of the abnormal size and book-to-market characteristic portfolio-adjusted return over trading days [0, +2] of the earnings announcement. The coefficient of interest is on $Post \times Covered \times Abs\_UE$, where *Abs_UE* is the decile ranking of the absolute value of unexpected earnings, calculated based on the IBES median analyst forecast.

I am interested in whether there is a decreased earnings announcement response to each unit of unexpected earnings, due to the pre-announcement price already incorporating some portion of the unexpected earnings. *Controls* in Model 1 include loss indicators, fourth fiscal quarter indicators, firm size, book-to-market, number of analysts, institutional ownership, stock

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which is consistent with greater price responses for small earnings surprises (Kinney et al. 2002). I interpret the decreased slope coefficient on absolute earnings surprises (β_7) as evidence that each additional unit of unexpected earnings results in a lower price reaction at the public announcement, because this surprise amount was already impounded into prices before the announcement.

17 My tests use June 30, 2014, as the *Post* date, because data sources 1 and 2 began selling data in 2014. Other data sources not directly used in this study also have rough start dates around this date, based on conversations with industry professionals. The choice of this *Post* date is also consistent with The Tabb Group estimates of the share of stock trading by quantitative-focused hedge funds, which significantly increased beginning in 2014 (Zuckerman and Hope 2017).

Table 2
Analysis of absolute earnings response coefficients

Model 1 Dependent variable: *Abs_AR[0, +2]*	**1**	**2**	**3**	**4**	**5**
Post	0.070	0.070	0.069	0.064	0.064
	(0.42)	(0.42)	(0.41)	(0.39)	(0.38)
Covered	0.525**	0.527**	0.524**	0.529**	0.534**
	(2.30)	(2.32)	(2.31)	(2.33)	(2.37)
Post×Abs_UE	0.018	0.018	0.018	0.019	0.019
	(0.76)	(0.75)	(0.75)	(0.77)	(0.78)
Covered×Abs_UE	0.067	0.071	0.067	0.068	0.065
	(1.46)	(1.53)	(1.48)	(1.49)	(1.45)
Post×Covered	0.365	0.361	0.382	0.431*	0.422*
	(1.51)	(1.51)	(1.61)	(1.84)	(1.78)
Post×Covered×Abs_UE	−0.030				
	(-0.63)				
Post×Covered[High_TAM]×Abs_UE		−0.009			
		(−0.16)			
Post×Covered[Low_TAM]×Abs_UE		−0.042			
		(−0.93)			
Post×Covered[High_MtoB]×Abs_UE			−0.050		
			(−1.03)		
Post×Covered[Low_MtoB]×Abs_UE			−0.017		
			(−0.36)		
Post×Covered[High_liq1]×Abs_UE				−0.124**	
				(−2.53)	
Post×Covered[Low_liq1]×Abs_UE				−0.014	
				(−0.29)	
Post×Covered[High_liq2]×Abs_UE					−0.127**
					(−2.41)
Post×Covered[Low_liq2]×Abs_UE					−0.001
					(−0.02)
Controls included?	Yes	Yes	Yes	Yes	Yes
Abs_UE and *Abs_UE×Controls* included?	Yes	Yes	Yes	Yes	Yes
F-stat *high* coefficient = *low* coefficient	–	1.15	1.08	8.36	10.73
Adjusted R^2	0.121	0.121	0.121	0.121	0.122
Observations	13,417	13,417	13,417	13,417	13,417

This table presents results of estimating Model 1:

$$\begin{aligned}Abs\_AR[0,+2] = {} & \beta_0+\beta_1 Post+\beta_2 Covered+\beta_3 (Post\times Covered)\\ & +\beta_4 Abs\_UE+\beta_5 (Abs\_UE\times Post)+\beta_6 (Abs\_UE\times Covered)\\ & +\boldsymbol{\beta_7} (Post\times Covered\times Abs\_UE)+\Sigma\beta_k Controls_k\\ & +\Sigma\beta_k (Abs\_UE\times Controls_k)+\Sigma\beta_j Industry_j\\ & +\Sigma\beta_j Industry_j\times Abs\_UE+\varepsilon,\end{aligned}$$

Observations are firm-quarters. *Controls* include *Fqtr_4*, *Market cap*, *BTM*, *Loss*, *Analysts*, *Abs_UE*, *Volatility*, *Instown*, *Beta*, *Persistence*, *Earn_volat*, and *Leverage. Controls*, *Abs_UE*, *Controls* interacted with *Abs_UE*, *Industry* fixed effects, and *Industry* fixed effects interacted with *Abs_UE* are included in all columns. Table A1 defines all variables. All continuous variables are winsorized at the 1% and 99% levels. Standard errors are clustered by *Industry* and quarter. ***, **, and * indicates significance at 1%, 5%, and 10% levels. Coefficients for *high* and *low* in shaded cells are significantly different from each other at the 1% level in Columns 4 and 5.

price volatility, leverage, beta, earnings persistence, and earnings volatility. *Abs_UE* × *Controls* are also included to control for the effect of these variables on the slope coefficient of *Abs_UE*.

Table 2 reports results of estimating Model 1. Columns 1 to 3 find that the coefficient on *Post* × *Covered* × *Abs_UE* is insignificant. Columns 4 and 5 find that the coefficient on *Post*×*Covered*×*Abs_UE* is significantly negative when *High_liq1* =1 or *High_liq2*=1. The reduction in the stock price reaction per unit of unexpected earnings is significantly more negative for firms with higher liquidity, in which sophisticated investors can more easily take positions. These firms rank in the bottom quartile of the Amihud (2002) illiquidity measure or the top quartile of dollar trading volume across all firms in my sample in the pre-period. In untabulated analyses, I address the concern that the *Controls* might affect the relation between positive and negative unexpected earnings and returns in a different way, so I interact the *Controls* with *UE* instead of *Abs_UE* and find substantively unchanged results. These results confirm that sophisticated investors seem to be trading in the pre-earnings announcement period and improving the incorporation into price of information related to the upcoming earnings announcement.

3.2.2 Long-run price informativeness. I define long-run price informativeness as the extent to which prices reflect the future earnings of the firm, where the future horizon is at least 1 year. Importantly, it is long-run price informativeness that can affect the manager's real decisions.[18] The long-run price informativeness measure is operationalized as the relation between current returns and future earnings (future earnings response coefficient, or FERC):

$$\begin{aligned} Ret_t = {} & \beta_0+\beta_1 Earn_{t-1}+\beta_2 Earn_t+\beta_3 Earn_{t+1}+\beta_4 Post+\beta_5 Covered \\ & +\beta_6(Post\times Covered)+\beta_7(Post\times Earn_{t-1})+\beta_8(Post\times Earn_t) \\ & +\beta_9(Post\times Earn_{t+1})+\beta_{10}(Covered\times Earn_{t-1})+\beta_{11}(Covered\times Earn_t) \\ & +\beta_{12}(Covered\times Earn_{t+1})+\beta_{13}(Post\times Covered\times Earn_{t-1}) \\ & +\beta_{14}(Post\times Covered\times Earn_t)+\boldsymbol{\beta_{15}}(Post\times Covered\times Earn_{t+1}) \\ & +\Sigma\beta_k Controls_k+\Sigma\beta_k Controls_k\times Earn_{t+1}+\Sigma\beta_j Industry_j \\ & +\Sigma\beta_j Industry_j\times Earn_{t+1}+\varepsilon. \end{aligned} \tag{2}$$

Model 2 tests for an increase in FERCs. The coefficient of interest is on $Post\times Covered\times Earn_{t+1}$. I test whether current returns reflect future earnings to a greater extent after the availability of alternative data. Following Israeli et al. (2017), each *Earn* variable is earnings before extraordinary items in the fiscal

[18] In Brandenburger and Polak (1996), managers are concerned with their share prices in a relevant window of uncertainty. If this window is short, in that the uncertainty related to their actions will be resolved in a week, then the information asymmetry between the market and managers will not be a problem in influencing managers' decisions. If alternative data shorten the window of uncertainty by a small amount, then there should be no effect on managers' actions. However, if the data increase long-run price informativeness, in that the window of uncertainty (the period of time that informational asymmetries persist) is significantly shortened due to revelation of managers' information that would not have been publicly disclosed for a long time, then managers' actions can be affected.

year indexed, deflated by market value at the beginning of the year. Kothari (1992) finds that this deflator results in lower bias of the estimated coefficient on earnings and higher explanatory power. Year $t-1$ earnings are included, to allow the regression to find the best representation of the market's expectation of current earnings using prior earnings (Lundholm and Myers 2002). Following Collins et al. (1994), *Controls* include future returns, size, loss indicators, asset growth, and the number of analysts. Future returns are included to account for the measurement error when using actual future earnings as a proxy for expected future earnings. Interactions of *Controls* with future earnings are also included, to control for the effect of these variables on the FERC. I include *Industry* and *Industry* fixed effects interacted with $Earn_{t+1}$ to control for industry-specific trends in returns and FERCs.

Table 3 reports results of estimating Model 2. In Column 1, the coefficient on $Post \times Covered \times Earn_{t+1}$ is positive and significant. Cross-sectional tests in Columns 2 to 5 find that this coefficient is generally significantly positive for firms for which sophisticated investors have the highest ex ante likelihood of uncovering information. I find that the coefficient on $Post \times Covered \times Earn_{t+1}$ is positive and significant when $High\_TAM = 1$ and $High\_liq1 = 1$. When $High\_MtoB = 1$, the coefficient on $Post \times Covered \times Earn_{t+1}$ is higher (significant at the 10% level) than the coefficient when $High\_MtoB = 0$. Sophisticated investors have higher incentives to uncover information for these firms, and liquidity also helps sophisticated investors take positions in these firms. The results suggest that price informativeness with respect to 1-year-ahead earnings has increased in the post-alternative data availability period.

3.3 The effect on managers' actions

The results in Section 3.2 suggest that prices have become more informative for firms covered by alternative data. The long-run price informativeness findings are that each dollar of future earnings is reflected to a greater extent in current returns. In this section, I first investigate whether the increase in long-run price informativeness changes the distribution of information rents, by focusing on managers' insider trading behavior. Then I investigate whether the change in the distribution between managers' private information and information available to informed market participants also has potential effects on the firm's investment efficiency.

3.3.1 Private information rents from insider trading. The following models test whether the insider trading behavior of managers changes after alternative data become available. In Model 3, the focus is on whether the directional magnitude of insider trading relates less to future earnings, conditional on insiders deciding to trade. In Model 4, the focus is on whether insiders' decision to purchase shares becomes less related to future earnings increases:

Table 3
Analysis of future earnings response coefficients

Model 2 Dependent variable: Ret_t	**1**	**2**	**3**	**4**	**5**
$Post\times Covered\times Earn_{t-1}$	−0.619*	−0.599*	−0.492	−0.544*	−0.539*
	(−1.92)	(−1.90)	(−1.44)	(−1.75)	(−1.70)
$Post\times Covered\times Earn_t$	0.098	0.159	0.171	−0.095	0.084
	(0.26)	(0.41)	(0.42)	(−0.25)	(0.24)
$Post\times Covered\times Earn_{t+1}$	0.511**				
	(2.30)				
$Post\times Covered[High\_TAM]\times Earn_{t+1}$		0.873**			
		(2.36)			
$Post\times Covered[Low\_TAM]\times Earn_{t+1}$		0.335*			
		(1.72)			
$Post\times Covered[High\_MtoB]\times Earn_{t+1}$			0.491		
			(1.30)		
$Post\times Covered[Low\_MtoB]\times Earn_{t+1}$			0.407		
			(1.62)		
$Post\times Covered[High\_liq1]\times Earn_{t+1}$				2.124***	
				(2.73)	
$Post\times Covered[Low\_liq1]\times Earn_{t+1}$				0.343	
				(1.41)	
$Post\times Covered[High\_liq2]\times Earn_{t+1}$					0.786
					(0.91)
$Post\times Covered[Low\_liq2]\times Earn_{t+1}$					0.411*
					(1.67)
Main effects and two-way interactions included?	Yes	Yes	Yes	Yes	Yes
Controls included?	Yes	Yes	Yes	Yes	Yes
$Controls_t\times Earn_{t+1}$ variables included?	Yes	Yes	Yes	Yes	Yes
F-stat *high* coefficient = *low* coefficient	–	6.65	3.64	9.83	2.09
Adjusted R^2	0.214	0.214	0.230	0.246	0.243
Observations	2,759	2,759	2,759	2,759	2,759

This table presents results of estimating Model 2:

$$\begin{aligned} Ret_t = {} & \beta_0+\beta_1 Earn_{t-1}+\beta_2 Earn_t+\beta_3 Earn_{t+1}+\beta_4 Post+\beta_5 Covered \\ & +\beta_6(Post\times Covered)+\beta_7(Post\times Earn_{t-1})+\beta_8(Post\times Earn_t) \\ & +\beta_9(Post\times Earn_{t+1})+\beta_{10}(Covered\times Earn_{t-1})+\beta_{11}(Covered\times Earn_t) \\ & +\beta_{12}(Covered\times Earn_{t+1})+\beta_{13}(Post\times Covered\times Earn_{t-1}) \\ & +\beta_{14}(Post\times Covered\times Earn_t)+\boldsymbol{\beta}_{15}(Post\times Covered\times Earn_{t+1}) \\ & +\Sigma\beta_k Controls_k+\Sigma\beta_k Controls_k\times Earn_{t+1}+\Sigma\beta_j Industry_j \\ & +\Sigma\beta_j Industry_j\times Earn_{t+1}+\varepsilon. \end{aligned}$$

Observations are firm-years. *Controls* include *Asset growth*, *log(market cap)*, Ret_{t+1}, $Loss_t$, and *Analysts*. *Controls*, *Controls* interacted with $Earn_{t+1}$, *Industry* fixed effects, and *Industry* fixed effects interacted with $Earn_{t+1}$ are included in all columns. Table A1 defines all variables. All continuous variables are winsorized at the 1% and 99% levels. Standard errors are clustered by *Industry*. ***, **, and * indicate significance at 1%, 5%, and 10% levels. Coefficients for *high* and *low* in shaded cells are significantly different from each other at the 1% level in Column 4, the 5% level in Column 2, and the 10% level in Column 3.

$$
\begin{aligned}
&\begin{array}{c} PurchaseRatio_{t+1} = \\ \text{or} \\ NetTrades_{t+1} \end{array} \beta_0+\beta_1 RetVar_{t+2}+\beta_2 EarnVar_{t+1}+\beta_3 EarnVar_{t+2} \\
&\quad +\beta_4 Post+\beta_5 Covered+\beta_6 (Post\times Covered) \\
&\quad +\beta_7 (Post\times RetVar_{t+2}) \\
&\quad +\beta_8 (Post\times EarnVar_{t+1})+\beta_9 (Post\times EarnVar_{t+2}) \\
&\quad +\beta_{10} (Covered\times RetVar_{t+2})+\beta_{11} (Covered\times EarnVar_{t+1}) \\
&\quad +\beta_{12} (Covered\times EarnVar_{t+2})+\beta_{13} (Post\times Covered\times RetVar_{t+2}) \\
&\quad +\beta_{14} (Post\times Covered\times EarnVar_{t+1}) \\
&\quad +\boldsymbol{\beta_{15}} (Post\times Covered\times EarnVar_{t+2}) \\
&\quad +\Sigma\beta_k Controls_k+\Sigma\beta_j Industry_j \\
&\quad +\Sigma\beta_j Industry_j\times EarnVar_{t+2}+\varepsilon,
\end{aligned} \tag{3}
$$

$$
\begin{aligned}
Pr(Purchase_{t+1}) = &\Phi(\beta_0+\beta_1 GoodRet_{t+2}+\beta_2 GoodROA_{t+1} \\
&+\boldsymbol{\beta_3} GoodROA_{t+2}+\Sigma\beta_k Controls_k+\varepsilon).
\end{aligned} \tag{4}
$$

The dependent variable in Model 3 is *PurchaseRatio* or *NetTrades*. *PurchaseRatio* is the ratio of insiders' open-market purchases to the sum of insiders' open-market purchases and sales in fiscal year $t+1$. *NetTrades* is the difference between insiders' open-market purchases and insiders' open-market sales, scaled by the sum of insiders' open-market purchases and sales in fiscal year $t+1$. I measure these variables in year $t+1$ to allow for managers' actions to occur after the measurement of *Post* in year t. *Controls* include the contemporaneous market-adjusted stock return, firm size, analyst coverage, book-to-market, the number of shares of restricted stock and stock options granted, and the number of stock options exercised, following Rozeff and Zaman (1998) and Piotroski and Roulstone (2005). *Controls* also include information environment variables, including institutional ownership and stock price volatility.

Table 4, Columns 1 to 5 report results of estimating Model 3 when the dependent variable is *PurchaseRatio*, $EarnVar_{t+1}$ is $\Delta Earn_{t+1}$, $EarnVar_{t+2}$ is $\Delta Earn_{t+2}$, and $RetVar_{t+2}$ is $AbnRet_{t+2}$. The coefficient of interest is on $Post\times Covered\times\Delta Earn_{t+2}$, as $\Delta Earn_{t+2}$ represents the unexpected portion of future earnings in the year after the measurement of *PurchaseRatio*, where year $t+1$ earnings represent the market expectation of year $t+2$ earnings. I also include interactions of $\Delta Earn_{t+1}$ and Ret_{t+2} with *Post*, *Covered*, and $Post\times Covered$, because alternative data availability can also affect insider trades' reflection of current earnings and future returns. Column 2 finds a negative and significant coefficient on $Post\times Covered\times\Delta Earn_{t+2}$. In cross-sectional analyses, I also find that the coefficient is significantly negative for subsets of firms. I find that the effect is concentrated in industries with a high total addressable market (Column 2) and high market-to-book firms (Column 3).

Table 4
Analysis of insider trades

Model 3	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Dependent variable:			$PurchaseRatio_{t+1}$					$NetTrades_{t+1}$		
$Post \times Covered \times EarnVar_{t+1}$	−0.074	−0.136	1.531	−0.542	0.478	−0.110	−0.108	−0.114	−0.097	−0.103
	(−0.11)	(−0.19)	(1.10)	(−0.19)	(0.17)	(−0.82)	(−0.81)	(−0.84)	(−0.70)	(−0.75)
$Post \times Covered \times RetVar_{t+2}$	−0.251	−0.241	−0.192	−0.275	−0.297	−0.206	−0.206	−0.204	−0.236	−0.230
	(−1.17)	(−1.12)	(−0.71)	(−1.11)	(−1.07)	(−1.40)	(−1.39)	(−1.41)	(−1.55)	(−1.50)
$Post \times Covered \times EarnVar_{t+2}$	−0.978**					−0.305**				
	(−2.14)					(−2.28)				
$Post \times Covered[High\_TAM] \times EarnVar_{t+2}$		−1.435***					−0.354**			
		(−2.64)					(−2.15)			
$Post \times Covered[Low\_TAM] \times EarnVar_{t+2}$		−0.681					−0.266*			
		(−1.48)					(−1.95)			
$Post \times Covered[High\_MtoB] \times EarnVar_{t+2}$			−1.345***					−0.288*		
			(−3.20)					(−1.88)		
$Post \times Covered[Low\_MtoB] \times EarnVar_{t+2}$			−1.547					−0.335**		
			(−0.70)					(−2.08)		
$Post \times Covered[High\_liq1] \times EarnVar_{t+2}$				−1.013					−0.431***	
				(−0.42)					(−3.59)	
$Post \times Covered[Low\_liq1] \times EarnVar_{t+2}$				−0.871*					−0.197	
				(−1.76)					(−1.17)	
$Post \times Covered[High\_liq2] \times EarnVar_{t+2}$					−0.643					−0.379***
					(−0.32)					(−3.14)
$Post \times Covered[Low\_liq2] \times EarnVar_{t+2}$					−0.939**					−0.226
					(−2.00)					(−1.31)
Controls included?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Main effects and two-way interactions included?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F-stat *high* coefficient = *low* coefficient	–	1.56	0.01	0.00	0.02	–	0.37	0.09	4.33	1.85
Adjusted R^2	0.315	0.315	0.317	0.324	0.309	0.174	0.173	0.173	0.173	0.173
Observations	847	847	847	847	847	2,075	2,075	2,075	2,075	2,075

This table presents results of estimating Model 3:

$$
\begin{aligned}
PurchaseRatio_{t+1} = \; & \beta_0+\beta_1 RetVar_{t+2}+\beta_2 EarnVar_{t+1}+\beta_3 EarnVar_{t+2}+\beta_4 Post+\beta_5 Covered+\beta_6(Post\times Covered)+\beta_7(Post\times RetVar_{t+2}) \\
\text{or} \quad & \\
NetTrades_{t+1} \quad & +\beta_8(Post\times EarnVar_{t+1})+\beta_9(Post\times EarnVar_{t+2})+\beta_{10}(Covered\times RetVar_{t+2})+\beta_{11}(Covered\times EarnVar_{t+1})+\beta_{12}(Covered\times EarnVar_{t+2}) \\
& +\beta_{13}(Post\times Covered\times RetVar_{t+2})+\beta_{14}(Post\times Covered\times EarnVar_{t+1})+\boldsymbol{\beta_{15}}(Post\times Covered\times EarnVar_{t+2})+\Sigma\beta_k Controls_k+\Sigma\beta_j Industry_j \\
& +\Sigma\beta_j Industry_j\times EarnVar_{t+2}+\varepsilon,
\end{aligned}
$$

Observations are firm-years. *Controls* include *Market cap*, *BTM*, *Loss*, *Analysts*, *Volatility*, $AbnRet_{t+1}$, *Grants*, and *Exercises*. *Controls*, *Industry* fixed effects, and *Industry* fixed effects interacted with $EarnVar_{t+2}$ are included in all columns. In Columns 1 to 5, the dependent variable is $PurchaseRatio_{t+1}$, $EarnVar_{t+1}$ is $\Delta Earn_{t+1}$, $EarnVar_{t+2}$ is $\Delta Earn_{t+2}$, and $RetVar_{t+2}$ is $AbnRet_{t+2}$. In Columns 5 to 8, the dependent variable is $NetTrades_{t+1}$, $EarnVar_{t+1}$ is $GoodROA_{t+1}$, $EarnVar_{t+2}$ is $GoodROA_{t+2}$, and $RetVar_{t+2}$ is $GoodRet_{t+2}$. Table A1 defines all variables. All continuous variables are winsorized at the 1% and 99% levels. Standard errors are clustered by *Industry*. ***, **, and * indicate significance at 1%, 5%, and 10% levels. Coefficients for *high* and *low* groups in shaded cells are significantly different from each other at the 5% level in Column 9.

In Columns 6 to 10, the dependent variable is $NetTrades_{t+1}$, $EarnVar_{t+1}$ is $GoodROA_{t+1}$, $EarnVar_{t+2}$ is $GoodROA_{t+2}$, and $RetVar_{t+2}$ is $GoodRet_{t+2}$. Following Piotroski and Roulstone (2005), I use these binary indicators of earnings and price increases and find similar results. The sample of firm-year observations is larger in Columns 6 to 10 because it includes firm-years with any insider trading activity, whereas Columns 1 to 5 include only firm-years with insider purchase activity. I find that the effect is concentrated in industries with a high total addressable market (Column 7) and more liquid firms (Columns 9 and 10). These firms are firms for which sophisticated investors have the highest incentives to uncover information and are the least constrained to trade on this information. In addition, results in Section 3.2.2 suggest that these same firms have the highest increases in long-run price informativeness. Untabulated analyses excluding routine trades, calculated following Cohen et al. (2012), find qualitatively similar results for both specifications of Model 3.

Model 4 tests whether the decision to trade changes after alternative data are available. I use a probit model to examine the observable outcome of the binary choice of insiders to purchase shares ($Purchase_{t+1}$) as a function of the change in future earnings. I am interested in whether the marginal probability effect of $GoodROA_{t+2}$ changes for covered firms relative to control firms. The marginal probability effect of $GoodROA_{t+2}$ is $\Phi(X_1'\beta)-\Phi(X_0'\beta)$, where $\Phi(X_1'\beta)$ is the value of the standard normal cumulative distribution function (CDF) at the independent variables X_1, setting $GoodROA_{t+2}=1$, and $\Phi(X_0'\beta)$ is the corresponding value, setting $GoodROA_{t+2}=0$. I average this marginal probability effect over all observations to obtain the estimated marginal probability effect. *Controls* follow those in Model 3. Following Lechner (2010), assuming common trends in the estimated coefficients of the latent model between the counterfactual of the Covered group without coverage and the control group, the difference-in-differences effect on the marginal probability effect of $GoodROA_{t+2}$ is $\Phi(X_1'\beta_{Covered,Post})-\Phi(X_0'\beta_{Covered,Post}) - [\Phi(X_1'(\beta_{Control,Post} - \beta_{Control,Pre}+\beta_{Covered,Pre})) - \Phi(X_0'(\beta_{Control,Post} - \beta_{Control,Pre}+\beta_{Covered,Pre}))]$.[19]

Figure 1 reports that the difference-in-differences effect on the marginal probability effect of $GoodROA_{t+2}$ on the probability of any insider purchase activity in year $t+1$ is -0.177. The predicted probability of insider purchase activity ahead of earnings increases is 17.7% lower for Covered companies after alternative data availability compared to the counterfactual outcome under no alternative data coverage. With a standard error of 0.094, calculated over

19 $\beta_{Covered,Post}$ denotes the estimated coefficients of the latent model estimated on the covered firms' post-period observations only. Identification relies on the assumption that the coefficients of the latent model of the post-period covered firms' insider trading choices, under the counterfactual that they are not covered by alternative data, are equivalent to $\beta_{Control,Post}$ - $\beta_{Control,Pre}+\beta_{Covered,Pre}$. Although this assumption is impossible to test directly, I alleviate concerns that this assumption is rejected by testing whether pre-period trends for $\beta_{Covered}$ and $\beta_{Control}$ are similar (in the Internet Appendix, I report separate β's estimated in each year in the pre-period, in each of the covered and control samples, and I show that the trends are not different across the two groups). In the Internet Appendix, I provide further details about the parallel trends assumption for this nonlinear model.

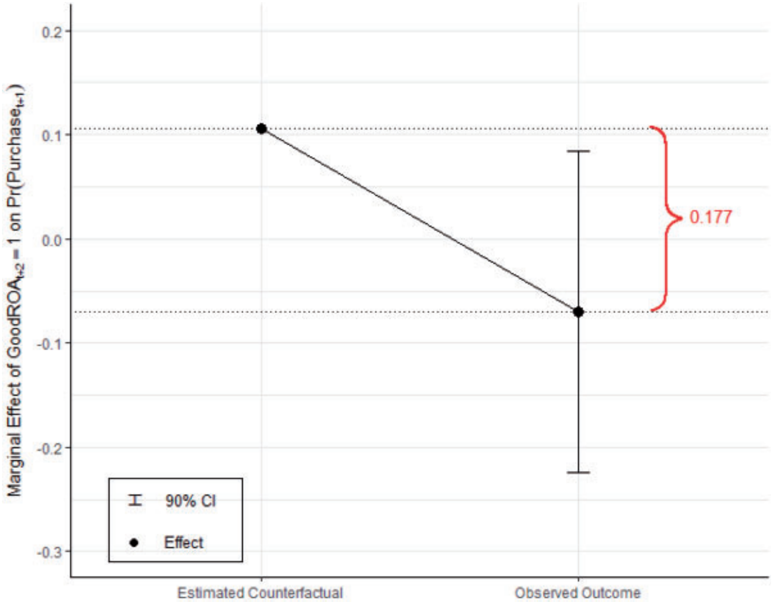

Figure 1
Effect of future earnings on the probability of insider purchases
This figure presents the estimated difference-in-differences effect on the marginal probability effect of $GoodROA_{t+2}=1$ on $Pr(Purchase_{t+1})$ in Model 4:

$$Pr(Purchase_{t+1}) = \Phi(\beta_0 + \beta_1 GoodRet_{t+2} + \beta_2 GoodROA_{t+1} + \beta_3 GoodROA_{t+2} + \Sigma\beta_k Controls_k + \varepsilon)$$

Observations are firm-years. *Controls* include: *Market cap*, *BTM*, *Loss*, *Analysts*, *Volatility*, $AbnRet_{t+1}$, *Grants*, and *Exercises*. Table A1 defines all variables. All continuous variables are winsorized at the 1% and 99% levels. The 90% confidence interval is plotted and calculated from standard errors based on 100,000 bootstrap draws, with replacement, of the same sample size as the number of firm-year observations, where *Covered*=1 and *Post*=1. See the Internet Appendix for further details.

100,000 bootstrap replications of the same sample size, this difference-in-differences effect is marginally significant. The symmetric effect on insider sales and earnings decreases is insignificant. Consistent with prior studies' findings that insider purchases contain more information than insider sales (e.g., Seyhun 1986; Lakonishok and Lee 2001), I find that alternative data availability disciplines managers' propensity to purchase shares, and I do not find the symmetric effect on managers' propensity to sell shares ahead of negative earnings news (untabulated). This result is also consistent with prior research's findings that insiders face more constraints and litigation risk on sales rather than buys (e.g., Section 16c of the U.S. Securities Exchange Act of 1934; Marin and Olivier 2008; Cohen et al. 2012).[20]

[20] Managers might use other complementary channels to exploit their private information. One such channel is disclosure. While tests of Models 3 and 4 find reduced information rent extraction through insider trading, I do not find the same effect when I examine disclosure. In untabulated analyses, I find no change in managers'

3.3.2 Investment efficiency. A second type of managerial action that can be disciplined by alternative data availability is related to a firm choice. More informative prices provide stronger incentives for the manager to make value-maximizing investment and divestment decisions. The misalignment of incentives is especially pertinent when investment opportunities are declining, because empire building and reputational concerns that prevent managers from discontinuing projects are more of a concern under these conditions (Kanodia et al. 1989; Boot 1992; Weisbach 1995).

My first investment efficiency test evaluates whether the firm's investment response to expanding and contracting investment opportunities changes after alternative data availability:

$$\begin{aligned}
log(\Delta I_{t+1}) = {} & \beta_0 + \beta_1 Post + \beta_2 Covered + \beta_3 (Post \times Covered) \\
& + \beta_4 IndustryRet_{t+1} + \beta_5 Neg + \beta_6 (Neg \times IndustryRet_{t+1}) \\
& + \beta_7 (Post \times IndustryRet_{t+1}) + \beta_8 (Post \times Neg) \\
& + \beta_9 (Post \times Neg \times IndustryRet_{t+1}) + \beta_{10} (Covered \times IndustryRet_{t+1}) \\
& + \beta_{11} (Covered \times Neg) + \beta_{12} (Covered \times Neg \times IndustryRet_{t+1}) \\
& + \boldsymbol{\beta_{13}} (Post \times Covered \times IndustryRet_{t+1}) + \beta_{14} (Post \times Covered \times Neg) \\
& + \boldsymbol{\beta_{15}} (Post \times Covered \times Neg \times IndustryRet_{t+1}) + \Sigma \beta_k Controls_k \\
& + \Sigma \beta_k Controls_k \times IndustryRet_{t+1} + \Sigma \beta_k Controls_k \times Neg \\
& + \Sigma \beta_k Controls_k \times Neg \times IndustryRet_{t+1} + \Sigma \beta_j Industry_j \\
& + \Sigma \beta_j Industry_j \times IndustryRet_{t+1} \\
& + \Sigma \beta_j Industry_j \times Neg \times IndustryRet_{t+1} + \varepsilon,
\end{aligned} \tag{5}$$

where the dependent variable $log(\Delta I_{t+1})$ is the log of the ratio of capital expenditures and R&D less sale of PP&E in year $t+1$ to capital expenditures and R&D in year t. The assumption is that the baseline level of investment, which is required to maintain existing assets and operations, is the capital expenditures and R&D from year t. Model 5 closely follows Bushman et al. (2011) and tests whether the sensitivity of investment response to improving and deteriorating investment opportunities changes for covered firms relative to control firms. The proxy for investment opportunities is industry returns in year t ($IndustryRet_t$), where the industry is measured based on product market similarities (Hoberg and Phillips 2010). *Controls* include book-to-market, firm

propensity to make the following types of announcements: guidance (both bundled at the earnings announcement and nonbundled guidance), product-related announcements, strategic alliances, and client announcements. It is unclear ex ante whether disclosure would increase after alternative data availability, because firms could disclose more information when prices are less informative (Frenkel et al. 2017; Balakrishnan et al. 2014).

size, and future returns. I include future returns to control for managers' market timing of investment, following Chen et al. (2007). I include *Industry* fixed effects, interacted with *Controls*, *Neg*, and *IndustryRet*, to allow the investment response to improving and deteriorating investment opportunities to vary by firm-specific and industry-specific characteristics.

I include an indicator variable, *Neg*, for deteriorating investment opportunities, to allow for an asymmetric change in the investment response to deteriorating investment opportunities relative to improving investment opportunities. Allowing for this asymmetric relation is important, because my hypotheses related to the disciplining effect of alternative data availability are stronger on the deteriorating opportunities side. If the availability of alternative data to market participants disciplines managers to invest more efficiently, then I should detect a positive difference-in-differences coefficient on the sensitivity of investment to deteriorating investment opportunities (i.e., positive $\beta_{13}+\beta_{15}$). This might constrain managers' tendencies to overinvest and mismanage assets in place when they should be decreasing investment (i.e., when investment opportunities are deteriorating). The optimality of investment choice with respect to the marginal value of capital is studied in the prior literature on q-theory (e.g., Tobin 1969; Hayashi 1982; Hubbard 1998). Alternative data might not increase the investment sensitivity to improving investment opportunities (i.e., no prediction for the sign of β_{13}), because the misalignment of managers' incentives with those of shareholders is not as strong under these conditions.

Table 5 reports results of estimating Model 5. In Column 1, I follow Model 5 but do not allow the response to *IndustryRet* to vary by whether *IndustryRet* is positive or negative (i.e., I omit all variables that are interacted with *Neg*, including any fixed effects interacted with *Neg).* In this specification, I find that the coefficient on $Post\times Covered\times IndustryRet_t$ is insignificant. In Column 2, I follow Model 5 exactly and allow the investment response to vary by whether industry returns are positive or negative, but I do not allow an industry-specific asymmetric response to deteriorating relative to improving investment opportunities (i.e., I omit $\Sigma\beta_j Industry_j\times Neg\times IndustryRet_t$). I find an insignificant coefficient on $Post\times Covered\times IndustryRet_t$ and a positive and significant coefficient on $Post\times Covered\times Neg\times IndustryRet_t$. The sum of these two coefficients is positive and significant (p = .007). Column 3 allows for an industry-specific asymmetric response to investment opportunities and finds similar, but stronger, results. The sum of the coefficients on $Post\times Covered\times IndustryRet_t$ and $Post\times Covered\times Neg\times IndustryRet_t$ remains positive and significant (p = .001).[21] Collectively, these results suggest that the

[21] In untabulated analyses, I find that the results from estimating Model 5 are robust to different specifications and sample transformations. I remove industry fixed effects, because of the concern that the fixed effects are measured using SIC codes, whereas *IndustryRet* is measured using TNIC sets. To address the concern that the dependent variable has slightly different distributions in the covered and control groups (i.e., the control firms have investment growth that is more concentrated at the tails of the distribution, relative to the covered group), I winsorize the dependent variable at 5% and 95%. My results are substantively unchanged from both of these robustness checks.

Table 5
Analysis of investment efficiency

Model 5 Dependent variable: $log(\Delta I_{t+1})$		**1**	**2**	**3**
(1)	*Post×Covered×IndustryRet*	−0.069	−0.683	−0.710
		(−0.24)	(−1.02)	(−1.03)
(2)	*Post×Covered×Neg×IndustryRet*		2.203**	2.218**
			(2.14)	(2.52)
(1) + (2)	Joint significance		**1.520****	**1.508****
			p = .007	**p = .001**
	Main effects, all two-way interactions, and all three-way interactions included?	Yes	Yes	Yes
	Average coefficient *Industry×IndustryRet*	0.276	0.392	0.598
	% positive and significant	78.0	80.0	94.0
	% negative and significant	22.0	20.0	6.0
	Average coefficient *Industry×Neg×IndustryRet*			0.305
	% positive and significant			60.2
	% negative and significant			39.8
	Industry, *Industry×IndustryRet* fixed effects included?	Yes	Yes	Yes
	Controls, *Controls×IndustryRet* included?	Yes	Yes	Yes
	Controls×Neg×IndustryRet included?	No	Yes	Yes
	Industry×Neg×IndustryRet fixed effects included?	No	No	Yes
	Adjusted R^2	0.109	0.117	0.110
	Observations	2,655	2,655	2,655

This table presents results of estimating Model 5:

$$\begin{aligned}
log(\Delta I_{t+1}) = {} & \beta_0 + \beta_1 Post + \beta_2 Covered + \beta_3 (Post \times Covered) \\
& + \beta_4 IndustryRet_{t+1} + \beta_5 Neg + \beta_6 (Neg \times IndustryRet_{t+1}) \\
& + \beta_7 (Post \times IndustryRet_{t+1}) + \beta_8 (Post \times Neg) \\
& + \beta_9 (Post \times Neg \times IndustryRet_{t+1}) + \beta_{10} (Covered \times IndustryRet_{t+1}) \\
& + \beta_{11} (Covered \times Neg) + \beta_{12} (Covered \times Neg \times IndustryRet_{t+1}) \\
& + \boldsymbol{\beta_{13}} (Post \times Covered \times IndustryRet_{t+1}) + \beta_{14} (Post \times Covered \times Neg) \\
& + \boldsymbol{\beta_{15}} (Post \times Covered \times Neg \times IndustryRet_{t+1}) + \Sigma \beta_k Controls_k \\
& + \Sigma \beta_k Controls_k \times IndustryRet_{t+1} + \Sigma \beta_k Controls_k \times Neg \\
& + \Sigma \beta_k Controls_k \times Neg \times IndustryRet_{t+1} + \Sigma \beta_j Industry_j \\
& + \Sigma \beta_j Industry_j \times IndustryRet_{t+1} + \Sigma \beta_j Industry_j \times Neg \times IndustryRet_{t+1} + \varepsilon,
\end{aligned}$$

Observations are firm-years. *Controls* include *Market cap*, *BTM*, and $AbnRet_{t+1}$. *Controls*, *Controls* interacted with *IndustryRet*, *Industry* fixed effects, and *Industry* fixed effects interacted with *IndustryRet* are included in all columns. Columns 2 and 3 include *Controls* interacted with *Neg* and *Controls* interacted with *Neg×IndustryRet*. Column 3 includes *Industry* fixed effects interacted with *Neg×IndustryRet*. Table A1 defines all variables. All continuous variables are winsorized at the 1% and 99% levels. Standard errors are clustered by *Industry*. ***, **, and * indicate significance at 1%, 5%, and 10% levels.

sensitivity of investment to deteriorating investment opportunities is higher after alternative data availability, consistent with the proposed disciplining effect of alternative data in constraining managers to make value-maximizing investments when prices reflect future earnings to a greater extent.[22]

---

[22] The results in this section are also consistent with managers learning from peer firm stock prices (Foucault and Fresard 2014). See Section 3.4 for further discussion.

My second test of investment efficiency further examines managers' divestment decisions and focuses on whether those decisions contribute more to shareholder value, after alternative data are available. I test for an increase in the abnormal returns at announcements of corporate downsizings and discontinued operations:

$$AR[0,\ +2]_{t+1} = \beta_0 + \beta_1 Post + \beta_2 Covered + \boldsymbol{\beta_3}(Post \times Covered) + \Sigma\beta_k Controls_k + \varepsilon, \tag{6}$$

where $AR[0,\ +2]_{t+1}$ is the abnormal return in the 3-day window after the announcement of discontinued operations or downsizings in year $t+1$. *Controls* follow those used in prior literature studying the abnormal returns to acquisition announcements and include firm size, market-to-book, leverage, the stock price runup prior to the announcement, and future returns (Masulis et al. 2007).[23]

Table 6 reports results of estimating Model 6. In Column 1, the coefficient on *Post×Covered* is positive but insignificant. In Column 2, I find that the coefficient is significantly positive for firms in industries with discretionary consumer spending. This result is consistent with the market assessing managers' discontinued operations and downsizings to be more value enhancing for the firm after alternative data are available.[24]

3.3.3 Sophisticated investors. A potential channel through which insider trading is reduced and investments become more efficient for firms covered by alternative data is a decrease in monitoring costs for sophisticated investors. In all the tests, the hypothesized effects on price informativeness and the disciplining of managers' decisions are stronger in firms in which sophisticated investors are able to trade and have incentives to uncover information. The disciplining channel proposes that managers are constrained by the knowledge that sophisticated investors can easily monitor their actions using alternative data. These sophisticated investors can then efficiently incorporate the information in alternative data about the value implications of managers' actions into stock prices. To provide evidence that monitoring costs of sophisticated investors have decreased, I document increased activity from sophisticated investors (see the Internet Appendix Section IA.2). In particular, I find that demand for borrowing in the equity lending market, or short sale activity, becomes more sensitive to yet-unannounced directional earnings.

23 Masulis et al. (2007) also include free cash flow as a control variable, but I do not include this variable because, while free cash flow provides resources for acquisitions, it does not provide resources to discontinue operations. I also include future returns, based on the argument that firms might time their investment and divestment decisions (Chen et al. 2007). For example, discontinuing operations might signal undervaluation; consistent with this market timing hypothesis, I find in tests of Model 6 a positive and significant coefficient on future returns (untabulated).

24 Another explanation for this result is that investors observe the manager's divestment activity with less noise, enabling them to better assess that an action is value maximizing for the firm.

Table 6
Analysis of abnormal returns at announcements of discontinued operations

Model 6 Dependent variable: $AR[0, +2]_{t+1}$	**1**	**2**	**3**	**4**	**5**
Post	−0.203	−0.645	−0.369	−1.439	−1.479
	(−0.46)	(−1.14)	(−0.34)	(−0.87)	(−0.78)
Covered	0.556**	−0.212	−0.483	0.442	0.639
	(2.09)	(−0.39)	(−1.05)	(0.68)	(1.62)
Post×Covered	0.703				
	(1.23)				
Post×Covered[High_TAM]		2.331***			
		(5.24)			
Post×Covered[Low_TAM]		1.018			
		(1.50)			
Post×Covered[High_MtoB]			0.434		
			(0.64)		
Post×Covered[Low_MtoB]			1.223		
			(1.19)		
Post×Covered[High_liq1]				0.252	
				(0.36)	
Post×Covered[Low_liq1]				2.109	
				(1.26)	
Post×Covered[High_liq2]					0.155
					(0.24)
Post×Covered[Low_liq2]					2.311
					(1.13)
Constant	−0.587	0.155	−0.014	−0.618	−0.250
	(−0.66)	(0.14)	(−0.02)	(−0.45)	(−0.19)
Controls included?	Yes	Yes	Yes	Yes	Yes
F-stat *high* coefficient = *low* coefficient	–	1.96	0.42	0.83	0.82
Adjusted R^2	0.005	0.004	0.006	0.004	0.004
Observations	1,575	1,575	1,575	1,575	1,575

This table presents results of estimating Model 6:

$$AR[0,\ +2]_{t+1} = \beta_0 + \beta_1 Post + \beta_2 Covered + \boldsymbol{\beta_3}(Post \times Covered) + \Sigma\beta_k Controls_k + \varepsilon,$$

Observations are firm-announcement days. *Controls* include $Size_t$, $MtoB_t$, $Leverage_t$, *FutureAbnRet*, and *AbnRet[-210, -11]*. Table A1 defines all variables. All continuous variables are winsorized at the 1% and 99% levels. Standard errors are clustered by *Industry* and quarter. ***, **, and * indicate significance at 1%, 5%, and 10% levels. Coefficients for *high* and *low* groups in shaded cells are not significantly different from each other at the 10% level in any column.

3.4 Additional analyses

The results in Section 3.3 are suggestive of alternative data availability disciplining managers to extract fewer information rents through personal trading and to make better investment and divestment decisions. In conjunction, both sets of results are consistent with alternative data revealing the manager's private information. The asymmetric firm response to declining investment opportunities is consistent with alternative data disciplining managers' empire building and excessive continuation of projects. To further support the disciplining channel, in untabulated analyses, I also find that this asymmetric investment efficiency result is concentrated in firms where managers have greater equity incentives. This cross-sectional finding is consistent with improved price informativeness providing improved incentives for managers to make efficient decisions. In addition, the increased excess returns to announcements of discontinued operations further supports the disciplining of the excessive continuation of projects. However, there are alternative explanations for these investment efficiency results. In this section, I acknowledge these explanations and attempt to pinpoint their plausibility.

Under this general class of alternative explanations, the manager has better information with which to make better investment decisions (e.g., Morck et al. 1990; Bushman and Smith 2001; Bond et al. 2012). He might acquire data on his own firm, learn from prices, acquire data on competitors, or improve technology to learn about customers. To the extent that these activities differ between covered and control groups and are correlated with the timing of the availability of alternative data, my results could be driven by these activities.

To address these alternative explanations, I validate my conjecture that firm-specific information in alternative data is not incremental to the manager's own information. I follow prior literature on managers' learning from stock price's aggregation of information (e.g., Dow and Gorton 1997; Dow and Rahi 2003; Chen et al. 2007; Bakke and Whited 2010). Chen et al. (2007) show that managers incorporate information from their own firms' stock prices into their investment decisions. If the investment-price sensitivity increases for covered firms after alternative data availability, then managers likely learn incremental information. This channel could be through their own stock prices, acquiring data directly, or improving technology to collect data similar to the data collected by third-party sources. The investment-price sensitivity can decrease if managers learn incremental information from *peer* stock prices, but not from their own stock price, and thus put less weight on their own stock price (Foucault and Fresard 2014). In their model, the manager observes a signal from his own stock price and his peer's stock price, and he weights both signals when choosing investment. When his peer's stock price becomes more informative (i.e., contains incremental information from the manager's perspective) for an exogenous reason, he optimally changes the weights on the signals such that he weights the peer stock price more and his own stock price less.

Tests of Model 5 in Section 3.3 find the investment response to peer firm stock prices increases after alternative data availability, when those stock prices are decreasing. In the next test, I separately assess the investment response to own firm returns:

$$\begin{aligned} log(\Delta I_{t+1}) = {} & \beta_0+\beta_1 Post+\beta_2 Covered+\beta_3 (Post\times Covered) \\ & +\beta_4 FirmRet_{t+1}+\beta_5 (Post\times FirmRet_{t+1}) \\ & +\beta_6 (Covered\times FirmRet_{t+1})+\boldsymbol{\beta_7} (Post\times Covered\times FirmRet_{t+1}) \\ & +\Sigma\beta_k Controls_k+\Sigma\beta_k Controls_k\times FirmRet_{t+1}+\Sigma\beta_j Industry_j \\ & +\Sigma\beta_j Industry_j\times FirmRet_{t+1}+\varepsilon. \end{aligned} \tag{7}$$

The dependent variable $log(\Delta I_{t+1})$ is the log of the ratio of the sum of capital expenditures and R&D less sale of PP&E in year $t+1$ to the sum of capital

Table 7
Analysis of investment-return sensitivity

	Model 7 Dependent variable: $log(\Delta I_{t+1})$
Post	-0.065^{***}
	(-3.24)
Covered	-0.023
	(-0.59)
Post×*Covered*	0.016
	(0.65)
Post×*FirmRet*	0.117^{***}
	(3.35)
Covered×*FirmRet*	0.075
	(0.97)
Post×*Covered*×*FirmRet*	-0.238^{***}
	(-3.26)
Average coefficient *Industry*×*FirmRet*	0.215
% positive and significant	71.7
% negative and significant	28.3
Industry, *Industry*×*FirmRet* fixed effects included?	Yes
Controls, *Controls*×*FirmRet* included?	Yes
Adjusted R^2	0.159
Observations	3,050

This table presents results of estimating Model 7:

$$log(\Delta I_{t+1}) = \beta_0 + \beta_1 Post + \beta_2 Covered + \beta_3 (Post \times Covered) + \beta_4 FirmRet_{t+1} + \beta_5 (Post \times FirmRet_{t+1})$$
$$+ \beta_6 (Covered \times FirmRet_{t+1}) + \boldsymbol{\beta_7} (Post \times Covered \times FirmRet_{t+1}) + \Sigma\beta_k Controls_k$$
$$+ \Sigma\beta_k Controls_k \times FirmRet_{t+1} + \Sigma\beta_j Industry_j + \Sigma\beta_j Industry_j \times FirmRet_{t+1} + \varepsilon.$$

Observations are firm-years. *Controls* include *Market cap*, *BTM*, and $AbnRet_{t+1}$. *Controls*, *Controls* interacted with *FirmRet*, *Industry* fixed effects, and *Industry* fixed effects interacted with *FirmRet* are included. Table A1 defines all variables. All continuous variables are winsorized at the 1% and 99% levels. Standard errors are clustered by *Industry*. ***, **, and * indicate significance at 1%, 5%, and 10% levels.

expenditures and R&D in year t. $FirmRet_t$ is firm returns measured in year t. *Controls* and fixed effects closely follow those in Model 5. I have no ex ante predictions about whether the firm learning effect is asymmetric with respect to expanding or deteriorating investment opportunities, so I do not allow the response to *FirmRet* to vary by whether industry returns are positive or negative.[25]

Table 7 reports results of estimating Model 7. In Column 1, I find a negative and significant coefficient on $Post \times Covered \times FirmRet_t$, consistent with managers relying less on their own stock prices. These results are consistent with managers learning no incremental information related to the additional information impounded into their own firms' stock prices. Data on their own firms' operations (e.g., daily sales) and growth potential were

[25] In untabulated analyses, for purposes of comparison to Model 5, when estimating Model 7 I also allow the investment response to *FirmRet* to vary by *Neg*. The sample shrinks by 419 observations because I require peer information from the Hoberg-Phillips data library to calculate *Neg*. I find that the difference-in-differences coefficient on the investment response to firm returns is larger in magnitude when investment opportunities are expanding (*Neg*=0). However, there is no hypothesized asymmetric learning effect predicting different investment responses to deteriorating and improving investment opportunities.

already accessible to the manager before these alternative data providers began to provide data to investors, and this information asymmetry is supported in the literature (Healy and Palepu 1993; Froot et al. 2017). It is unlikely that managers learn incremental information about their own firm from these data providers. Nor are the results consistent with managers learning more from data collection on their own customers compared to the pre-alternative data availability period. However, the decreased weight on own firm stock prices suggests that the remaining learning channel is plausible: learning from peers, whether through directly acquiring data on competitors or learning from peers' stock prices.

While I acknowledge that I cannot rule out the learning-from-peers channel, the evidence I have shown is consistent with managers being disciplined.[26] The reduced rent extraction through personal trading is consistent with the incorporation of alternative data into prices disciplining managers' personal trading activities. Furthermore, prior literature has proposed multiple agency problems that can affect managers' investment and divestment decisions (Jensen 1986; Kanodia et al. 1989; Boot 1992; Weisbach 1995). Consistent with the alleviation of these agency problems, I find an asymmetric ability of alternative data availability to discipline managers' investment and divestment decisions when investment opportunities are contracting. There is a misalignment of incentives when investment opportunities are deteriorating, such that revelation of alternative data to the market results in better downsizing decisions, as evidenced by the increased abnormal returns to announcements of discontinued operations.

4. Conclusion

Economic agents adjust their activities based on changes in price, a property Hayek (1945) argues is the essence of a competitive price system. Tobin (1984, p. 10) discusses the important role of the price system in functional efficiency, which is "the services the financial industries perform for the economy as a whole." In this study, I document improved price informativeness and a disciplining effect on managers, using a decrease in information acquisition costs that is exogenous to the firm's managers. My empirical results provide evidence that, following the availability of alternative data, such as consumer transactions and satellite images, price informativeness improves. Future earnings are incorporated more quickly and completely into current returns. In cross-sectional tests that find the price informativeness results are concentrated in firms for which sophisticated investors have the highest incentives to uncover

[26] Prior literature has found that learning from market prices contributes little to market resource allocation (David et al. 2016). My results are consistent both with managers learning from the prices of peers and managers learning directly from alternative data on peers. The high prices of the alternative data sets prevent managers from acquiring these data directly, and data providers I have spoken to do not want to sell to corporate managers. However, I cannot completely rule out this direct channel, especially for private companies (not in my sample), such as Uber and Lyft, acquiring data on each other from a third-party data provider (Isaac and Lohr 2017).

information, I provide further evidence that the observed effect is related to sophisticated investors acquiring alternative data sets.

One effect of this increase in long-run price informativeness on managers' actions is the reduced extraction of information rents to increase their personal wealth. Managers have less of an opportunity to trade on their private information about future earnings when prices reflect those future earnings to a greater extent. I find that directional insider trading activity has a reduced relation with future earnings. Furthermore, I find that insiders are less likely to purchase shares ahead of 1-year-ahead positive earnings news.

The second effect on managers' actions I document is the impact on real investment decisions. Alternative data presumably reveal information about the current business and future growth opportunities of the firm. I find that, relative to those of a group of control firms, the investment responses of firms covered by alternative data are more sensitive to declining investment opportunities. Thus, the increase in investment efficiency seems to be associated with reduced agency problems. Agency problems are especially relevant when investment opportunities are deteriorating, and in further evidence consistent with this hypothesis, I find increased abnormal returns to announcements of discontinued operations.

I contribute to the literature linking price informativeness with managerial decisions by studying a change in price informativeness that is exogenous to managers' choices. I also contribute to the literature on technological innovations, information acquisition costs, and capital market outcomes. My study focuses on two data sets and their coverage of consumer-focused firms, which I hope will stimulate further investigation into other types of alternative data. While this study considers certain types of data arising from the "big data" revolution as "alternative data," as these types of data become mainstream, they may not be called "alternative" in the future. In the short period since investment professionals have been using these types of data, observable capital market effects and effects on managerial behavior already have been experienced.

Appendix

Table A1
Variable descriptions

Variable name	Description	Source
$\Delta Earn_{t+1}$	Change in 1-year-ahead earnings, calculated as Compustat IB in the next fiscal year less Compustat IB in the current fiscal year, all scaled by Compustat AT in the current fiscal year	Compustat
$\Delta Earn_{t+2}$	Change in 2-year-ahead earnings, calculated as Compustat IB in year $t+2$ less Compustat IB in the next fiscal year, all scaled by Compustat AT in the current fiscal year	Compustat
$AbnRet_{t+1}$	Market-adjusted buy-and-hold returns calculated over year $t+1$	CRSP
$AbnRet_{t+2}$	Market-adjusted buy-and-hold returns calculated over year $t+2$	CRSP
Abs_AR[0, +2]	Within-year decile ranking of the absolute value of abnormal returns in the [0, +2] window relative to the earnings announcement. Abnormal returns are calculated as the firm's buy-and-hold returns in the [0, +2] window less the equally weighted portfolio returns of firms in the same size and book-to-market quintile over the same window	CRSP, Compustat
Abs_UE	Within-year decile ranking of the absolute value of unexpected earnings (UE). The decile ranking output is from 0 to 9. It is the within-year decile ranking of the absolute value of UE calculated from IBES, if available, and the within-year decile ranking of the absolute value of UE calculated as a seasonal random walk, if IBES information is unavailable. UE from IBES is the actual EPS less the median IBES forecast, scaled by price at the fiscal quarter end. UE from the seasonal random walk model is current quarter EPS less four-quarters-prior EPS, scaled by prior-year fiscal quarter end price	IBES, CRSP, Compustat
Analysts	Logarithm of 1 plus the number of analysts	IBES
$AR[0, +2]_{t+1}$	Abnormal returns in the [0, +2] window relative to the announcement date of discontinued operations (Capital IQ keyDevEventTypeId = 21) in year $t+1$. Abnormal returns are calculated as the firm's buy-and-hold returns in the [0, +2] window less the equally weighted portfolio returns of firms in the same size and book-to-market quintile over the same window, multiplied by 100	CRSP, Compustat, Capital IQ
Asset growth	Growth rate of total assets, calculated as current Compustat AT divided by previous fiscal year Compustat AT, minus 1	Compustat
Beta	Stock market beta, calculated over trading days [-252, -5] relative to the earnings announcement	CRSP
BTM	Book-to-market ratio, calculated as Compustat CEQ divided by market value	CRSP, Compustat
Covered	An indicator variable equal to 1 for firms covered by alternative data providers. See Section 3.1 for details on the sample construction of the covered group	Compustat, Hoberg-Phillips data library
Earn_volat	Earnings volatility, calculated as the standard deviation of the seasonal difference in EPS, calculated for the trailing 4 years	Compustat
$Earn_{t-1}$	Prior-year earnings, calculated as Compustat IB in the previous fiscal year, scaled by market value at the beginning of the previous fiscal year	Compustat
$Earn_t$	Contemporaneous earnings, calculated as Compustat IB in the current fiscal year, scaled by market value at the beginning of the current fiscal year	Compustat
$Earn_{t+1}$	Future earnings, calculated as Compustat IB in the next fiscal year, scaled by market value at the beginning of the next fiscal year	Compustat
Exercises	Logarithm of the ratio of the number of stock options exercised by insiders in the next fiscal year to total shares outstanding	Thomson
FirmRet	Logarithm of 1 plus returns in year $t+1$	CRSP
Fqtr_4	An indicator variable equal to 1 if the fiscal quarter is the fourth fiscal quarter	Compustat

Table A1
(Cont.)

Variable name	Description	Source
FutureAbnRet	Market-adjusted buy-and-hold returns calculated over the 3 years beginning at the end of year $t+1$. If 3-year returns are unavailable, 2-year returns are used, and if 2-year returns are unavailable, 1-year returns are used	CRSP
$GoodRet_{t+2}$	An indicator variable equal to 1 if $AbnRet_{t+2}$ is positive	CRSP
$GoodROA_{t+1}$	An indicator variable equal to 1 if $\Delta Earn_{t+1}$ is positive	Compustat
$GoodROA_{t+2}$	An indicator variable equal to 1 if $\Delta Earn_{t+2}$ is positive	Compustat
Grants	Logarithm of the ratio of the number of shares of restricted stock and options granted to insiders in the next fiscal year to total shares outstanding	Thomson
High_liq1	An indicator variable equal to 1 if the firm's Amihud (2002) illiquidity measure is in the bottom quartile of the sample. The Amihud (2002) illiquidity measure is calculated as the average daily ratio of absolute stock return to dollar trading volume, where the average is computed over the trading days in the last month of the last pre-period quarter. *Covered[High_liq1]* is an indicator variable equal to 1 if *Covered*=1 and *High_liq1*=1	CRSP
High_liq2	An indicator variable equal to 1 if the firm's average daily dollar trading volume in the last month of the last pre-period quarter is in the top quartile of the sample. *Covered[High_liq2]* is an indicator variable equal to 1 if *Covered*=1 and *High_liq2*=1	CRSP
High_MtoB	Indicator variable equal to 1 if the firm's market-to-book ratio, calculated as market value divided by Compustat CEQ, is above the median market-to-book ratio. The market-to-book ratio is calculated as of the last pre-period quarter. *Covered[High_MtoB]* is an indicator variable set to 1 if *Covered*=1 and *High_MtoB*=1	CRSP, Compustat
High_TAM	Indicator variable equal to 1 if the firm is in an industry with a high total addressable market (TAM), defined as industries with discretionary consumer spending. Specifically, these are firms in the restaurant, apparel, automotive, and travel industries, where the industry classification is as of the last pre-period quarter. *Covered[High_TAM]* is an indicator variable equal to 1 if *Covered*=1 and *High_TAM*=1	Compustat
Industry	Two-digit SIC industry	CRSP
IndustryRet	Logarithm of 1 plus the industry returns in year $t+1$, where the industry returns are the average stock return for firms in the same product market space, based on the Hoberg and Phillips (2010) text-based industry classifications	Hoberg-Phillips data library, CRSP
Instown	Fraction of shares held by institutional investors, calculated at the most recent file date between 100 days prior to the earnings announcement date and the earnings announcement date	WhaleWisdom
Leverage	The ratio of total liabilities (Compustat LTQ or LT) to total equity (Compustat SEQQ or SEQ if available, ATQ-LTQ or AT-LT if SEQQ or SEQ are not available), measured at the end of the fiscal year or quarter	Compustat
log(market cap)	Logarithm of beginning of year market cap, measured in millions and calculated as Compustat PRCC_F*CSHO at the end of the previous fiscal year	Compustat
$log(\Delta I_{t+1})$	Logarithm of the ratio of investment in year $t+1$ to capital expenditures and R&D in year t, where capital expenditures and R&D are calculated as the sum of Compustat CAPX and Compustat XRD. Investment in year $t+1$ includes sale of PP&E and is the sum of Compustat CAPX and Compustat XRD less Compustat SPPE. If CAPX, XRD, or SPPE are missing, then the missing variable equals 0. If both CAPX and XRD are missing or 0 in year t, then $log(\Delta I_{t+1})$ is set as missing	Compustat

Table A1
(Cont.)

Variable name	Description	Source
Loss	An indicator variable equal to 1 if the quarterly EPS is negative. EPS is the actual EPS from IBES and Compustat EPSFXQ if the IBES EPS is unavailable	IBES, Compustat
$Loss_t$	An indicator variable equal to 1 if annual earnings are negative. Earnings are defined as in Compustat IB	Compustat
Low_liq1	Binary reciprocal of *High_liq1*. *Covered[Low_liq1]* is an indicator variable equal to 1 if *Covered* = 1 and *Low_liq1* = 1	CRSP
Low_liq2	Binary reciprocal of *High_liq2*. *Covered[Low_liq2]* is an indicator variable equal to 1 if *Covered* = 1 and *Low_liq2* = 1	CRSP
Low_MtoB	Binary reciprocal of *High_MtoB*. *Covered[Low_MtoB]* is an indicator variable equal to 1 if *Covered* = 1 and *Low_MtoB* = 1	CRSP, Compustat
Low_TAM	Binary reciprocal of *High_TAM*. *Covered[Low_TAM]* is an indicator variable equal to 1 if *Covered* = 1 and *Low_TAM* = 1	Compustat
Market cap	Quarter end market cap in millions, calculated as Compustat PRCCQ*CSHOQ. If Compustat variables are missing, then the variable is calculated as CRSP abs(PRC)*SHROUT/1,000	CRSP, Compustat
MtoB	Market-to-book ratio, calculated as market value divided by Compustat CEQ	CRSP, Compustat
Neg	An indicator variable equal to 1 if *IndustryRet* is negative	Hoberg-Phillips data library, CRSP
NetTrades	The number of shares purchased by insiders less the number of shares sold by insiders, scaled by the sum of the number of shares purchased and the number of shares sold by insiders in year $t+1$. Transactions are restricted to open-market transactions, and firm-years without insider purchase or sale activity are excluded	Thomson
Persistence	Earnings persistence, calculated as the AR(1) coefficient of regressing current earnings on prior-year earnings in the same quarter, calculated over the trailing 4 years.	Compustat
Post	An indicator variable equal to 1 if the end of the fiscal quarter is June 30, 2014 or later	Compustat
Purchase	An indicator variable equal to 1 if insiders purchased any shares in any open-market transactions in year $t+1$	Thomson
PurchaseRatio	The number of shares purchased by insiders, scaled by the sum of the number of shares purchased and the number of shares sold by insiders in year $t+1$. Transactions are restricted to open-market transactions, and firm-years without insider purchase activity are excluded	Thomson
Ret_t	Buy-and-hold returns calculated over the current fiscal year	CRSP
Ret_{t+1}	Buy-and-hold returns calculated over year $t+1$	CRSP
Size	Logarithm of total assets (Compustat AT)	Compustat
UE	Within-year decile ranking of unexpected earnings (UE). The decile ranking output is from 0 to 9. It is the within-year decile ranking of UE calculated from IBES, if available, and the within-year decile ranking of UE calculated as a seasonal random walk, if IBES information is unavailable. UE from IBES is the actual EPS less the median IBES forecast, scaled by price at the fiscal quarter end. UE from the seasonal random walk model is current quarter EPS less four-quarters-prior EPS, scaled by prior-year fiscal quarter end price	IBES, CRSP, Compustat
Volatility	Stock return volatility, calculated as the standard deviation of stock returns over the previous quarter or year, multiplied by $\sqrt{252}$ (annualized standard deviation of returns). In the firm-year models, *Volatility* is calculated over the previous quarter, and in the firm-year models, *Volatility* is calculated over the previous year	CRSP

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